

Phylogenetics

Introduction to Bayesian
inference and MCMC

RL-V3 MPP

Rachel Warnock

08.04.26



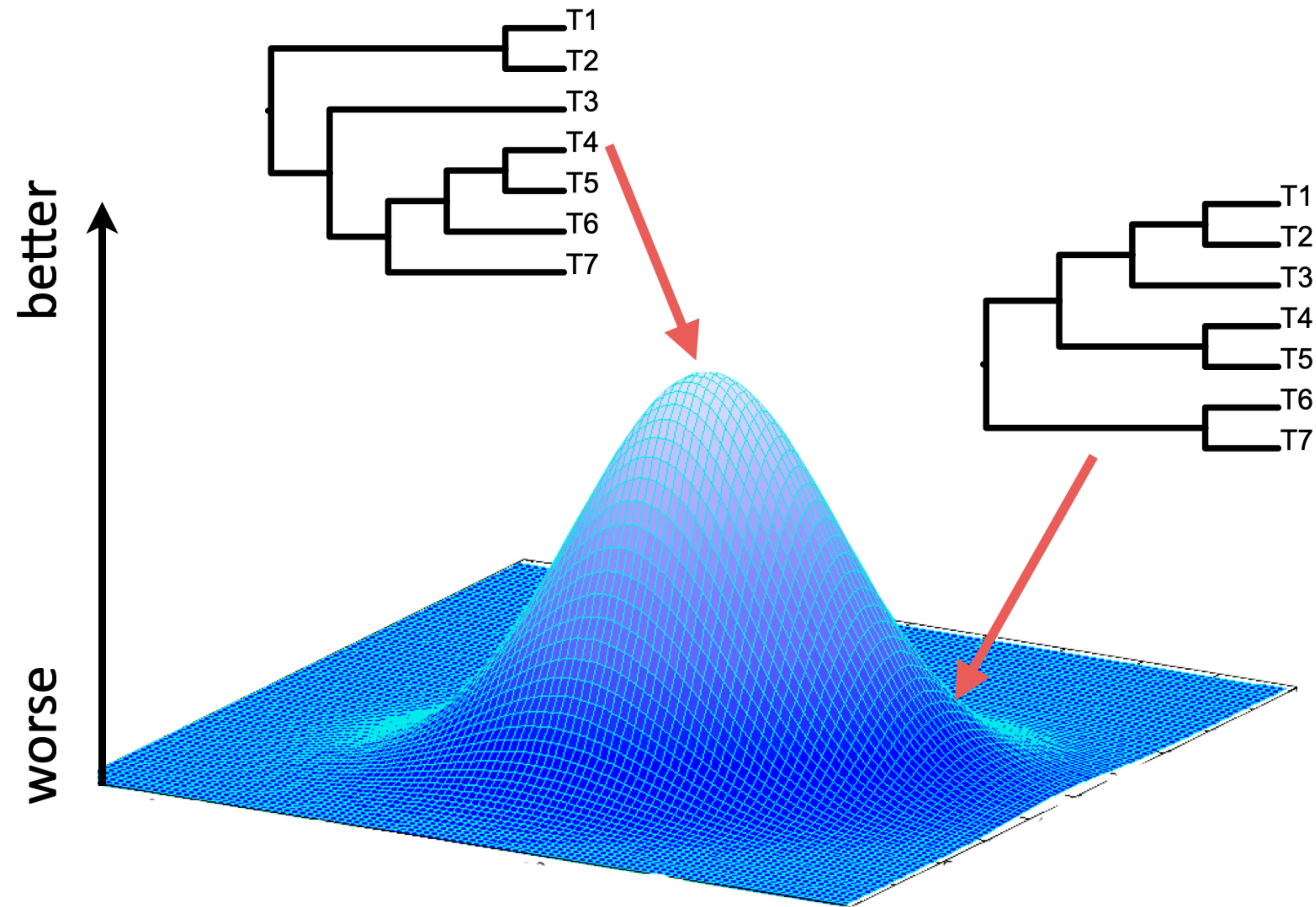
Objectives

- Bayesian inference
- MCMC
- MrBayes



Recap

How do we find the 'best' tree?



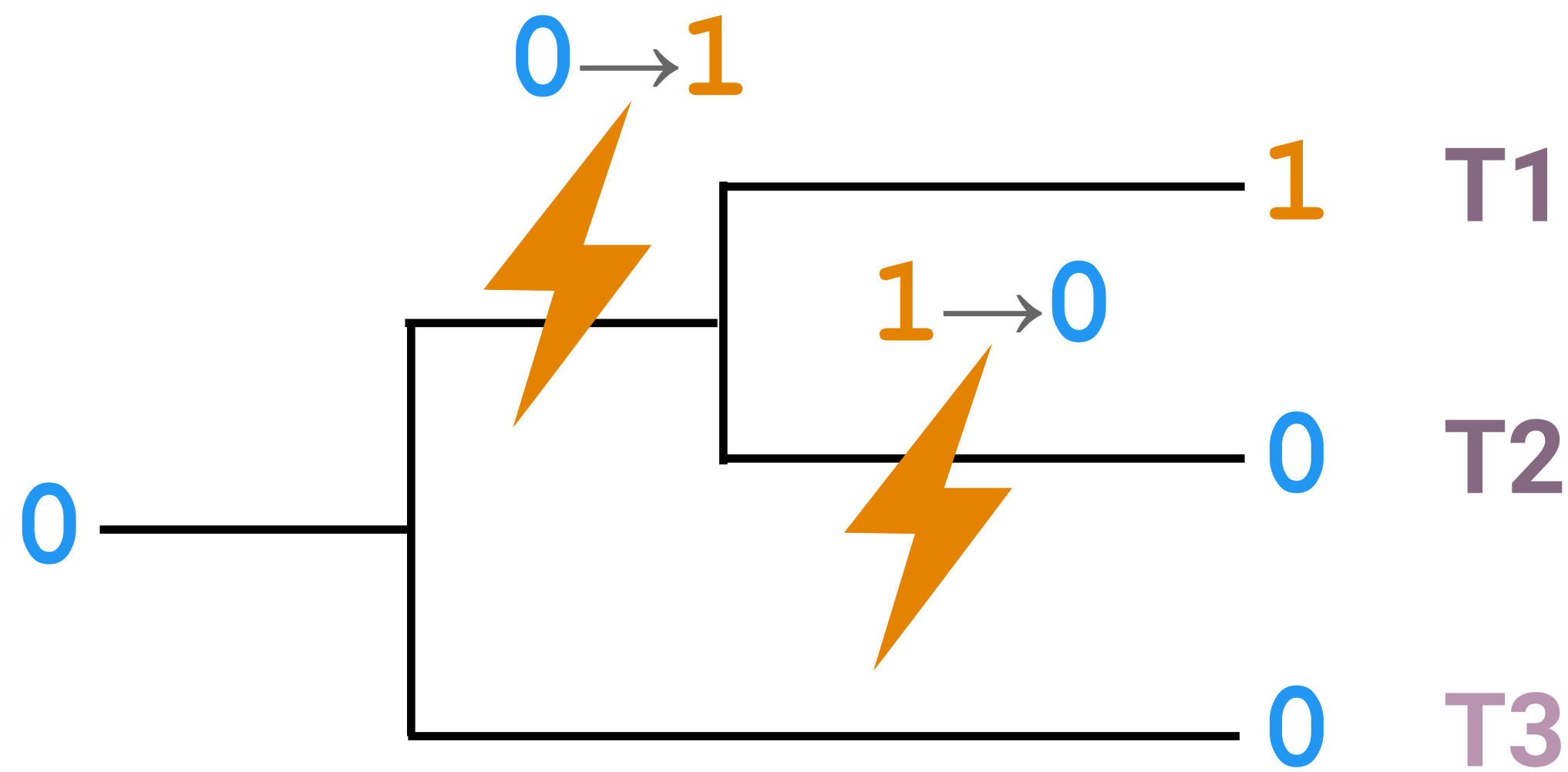
It depends how you measure 'best'

Method	Criterion (tree score)
Maximum parsimony	Minimum number of changes
Maximum likelihood	Likelihood score (probability), optimised over branch lengths and model parameters
Bayesian inference	Posterior probability, integrating over branch lengths and model parameters

Both maximum likelihood and Bayesian inference are model-based approaches

Note these are not the only approaches to tree-building but they are the most widely used

Convergence and parsimony



Hypothetical tree showing multiple transitions at the same character

Parsimony will always favour the tree with the smallest number of changes

The method does not account for multiple transitions (or “hits”), e.g.,
 $0 \rightarrow 1 \rightarrow 0$

We can only invoke convergent evolution *ad hoc* after inference

Recap: parsimony

Parsimony does not make **explicit** assumptions about the evolutionary process (although it makes **implicit** assumptions)

It has been demonstrated that in some scenarios parsimony is **statistically inconsistent**. The issue is known as **long branch attraction**

Model-based approaches on the other hand make **explicit** assumptions about evolutionary processes (+ have a wider range of applications in paleobio)

It depends how you measure 'best'

Method	Criterion (tree score)
Maximum parsimony	Minimum number of changes
Maximum likelihood	Likelihood score (probability), optimised over branch lengths and model parameters
Bayesian inference	Posterior probability, integrating over branch lengths and model parameters

Both maximum likelihood and Bayesian inference are model-based approaches

Note these are not the only approaches to tree-building but they are the most widely used

Bayesian tree inference

Bayes' theorem

$$\Pr(\text{model} \mid \text{data}) = \frac{\Pr(\text{data} \mid \text{model}) \Pr(\text{model})}{\Pr(\text{data})}$$

Bayes' theorem

Likelihood

The probability of the data given the model assumptions and parameter values

$$\Pr(\text{model} \mid \text{data}) = \frac{\Pr(\text{data} \mid \text{model}) \Pr(\text{model})}{\Pr(\text{data})}$$

Bayes' theorem

Priors

This represents our prior knowledge of the model parameters

$$\Pr(\text{model} \mid \text{data}) = \frac{\Pr(\text{data} \mid \text{model}) \Pr(\text{model})}{\Pr(\text{data})}$$

Bayes' theorem

$$\Pr(\text{model} \mid \text{data}) = \frac{\Pr(\text{data} \mid \text{model}) \Pr(\text{model})}{\Pr(\text{data})}$$

Marginal probability

The probability of the data, given all possible parameter values. Can be thought of as a normalising constant

Bayes' theorem

Reflects our combined knowledge based on the likelihood and the priors

posterior

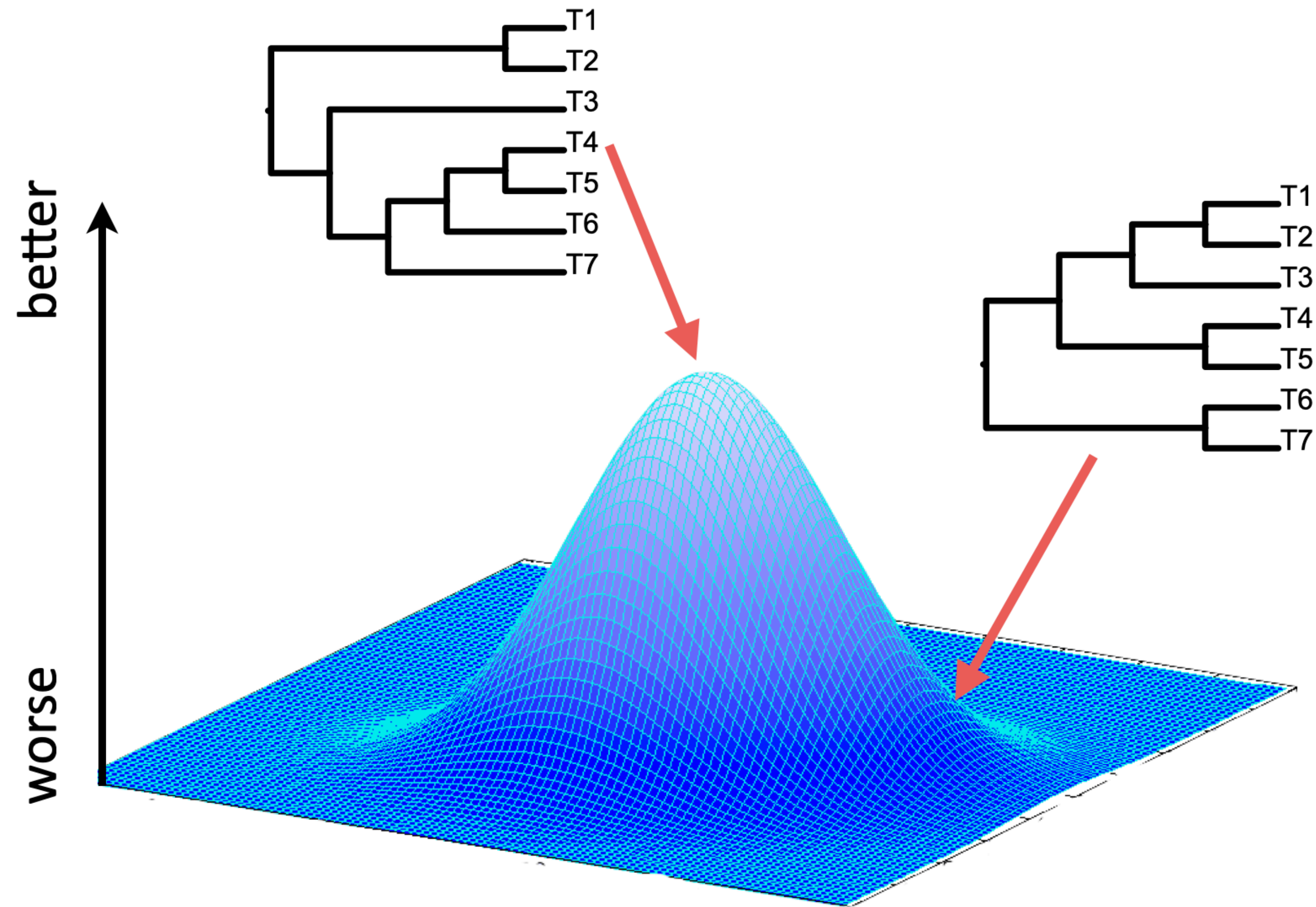
$$\text{Pr}(\text{model} \mid \text{data}) = \frac{\text{Pr}(\text{data} \mid \text{model}) \text{Pr}(\text{model})}{\text{Pr}(\text{data})}$$

Bayes' theorem

$$\Pr(\text{model} \mid \text{data}) \propto \Pr(\text{data} \mid \text{model}) \Pr(\text{model})$$

The posterior is proportional to the product of the prior and the likelihood

How do we find the 'best' tree?



It depends how you measure 'best'

Method	Criterion (tree score)
Maximum parsimony	Minimum number of changes
Maximum likelihood	Likelihood score (probability), optimised over branch lengths and model parameters
Bayesian inference	Posterior probability, integrating over branch lengths and model parameters

Both maximum likelihood and Bayesian inference are model-based approaches

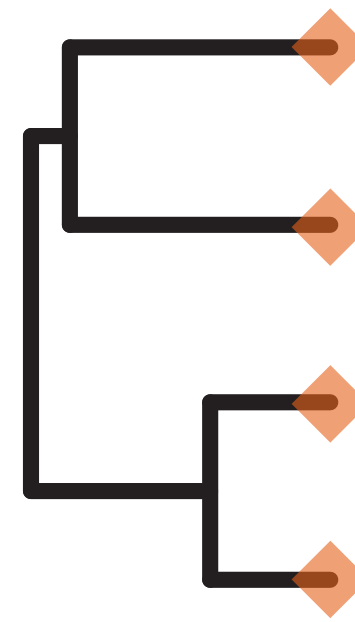
Note these are not the only approaches to tree-building but they are the most widely used

Components used to infer trees

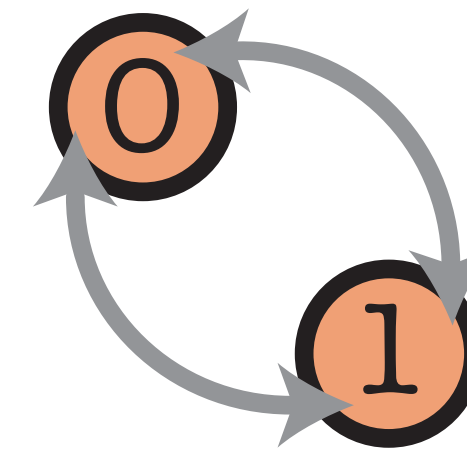
without considering time

0101...
1101...
0100...

data
sequences or
characters



tree
topology and
branch lengths



substitution
model

Bayesian tree inference

$$\begin{array}{c} \text{posterior} \\ \boxed{} \end{array}
 \quad
 P\left(\begin{array}{c} \text{tree} \\ \text{0} \rightarrow \text{1} \end{array} \mid \begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \right)
 \quad
 =
 \quad
 \begin{array}{cc} \text{likelihood} & \text{priors} \\ \boxed{} & \boxed{} \end{array}
 \quad
 \frac{P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \begin{array}{c} \text{tree} \\ \text{0} \rightarrow \text{1} \end{array} \right) P\left(\begin{array}{c} \text{tree} \\ \text{0} \rightarrow \text{1} \end{array} \right)}{P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \right)}
 \quad
 \begin{array}{c} \text{marginal probability} \\ \boxed{} \end{array}$$

Introduction to MCMC

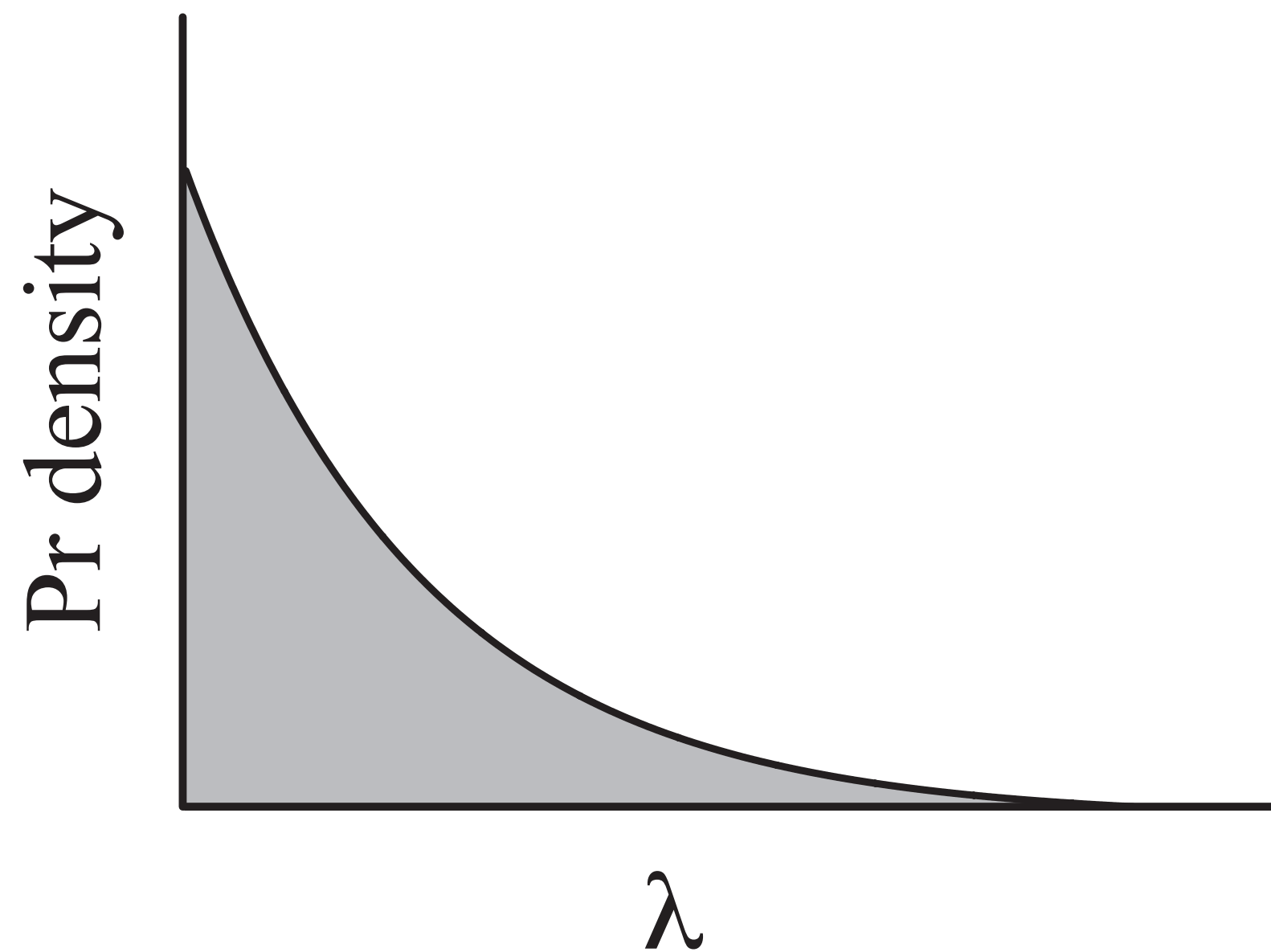
Probabilities vs. probability densities

In phylogenetics, probabilities are not normally discrete (i.e., represented by a single value)

We're often dealing with a lot of uncertainty and typically work with **probability densities**

Probability densities introduce some complexity

Probabilities vs. probability densities



λ is drawn from an exponential distribution with mean δ

The x-axis represents the value of our parameter λ

The y-axis does have a value but it is not so easily interpretable

The distribution height reflects the relative probability of a given range of values

Bayesian tree inference

$$\begin{array}{c} \text{posterior} \\ \boxed{} \end{array}
 \quad
 P\left(\begin{array}{c} \text{tree} \\ \text{0} \rightarrow \text{1} \end{array} \mid \begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \right)
 \quad
 =
 \quad
 \begin{array}{c} \text{likelihood} \qquad \qquad \text{priors} \\ \boxed{} \qquad \boxed{} \end{array}
 \quad
 \frac{P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \begin{array}{c} \text{tree} \\ \text{0} \rightarrow \text{1} \end{array} \right) P\left(\begin{array}{c} \text{tree} \\ \text{0} \rightarrow \text{1} \end{array} \right)}{P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \right)}
 \quad
 \begin{array}{c} \text{marginal probability} \\ \boxed{} \end{array}$$

Bayesian tree inference

$$= \frac{P\left(\begin{matrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{matrix} \middle| \begin{matrix} \text{tree} \\ \text{0} \rightarrow \text{1} \end{matrix}\right) P\left(\begin{matrix} \text{tree} \\ \text{0} \rightarrow \text{1} \end{matrix}\right)}{\int P\left(\begin{matrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{matrix} \middle| \begin{matrix} \text{tree} \\ \text{0} \rightarrow \text{1} \end{matrix}\right) P\left(\begin{matrix} \text{tree} \\ \text{0} \rightarrow \text{1} \end{matrix}\right) d\begin{matrix} \text{tree} \\ \text{0} \rightarrow \text{1} \end{matrix}}$$

this part is incredibly
difficult to calculate!

What is Markov chain Monte Carlo (MCMC)?

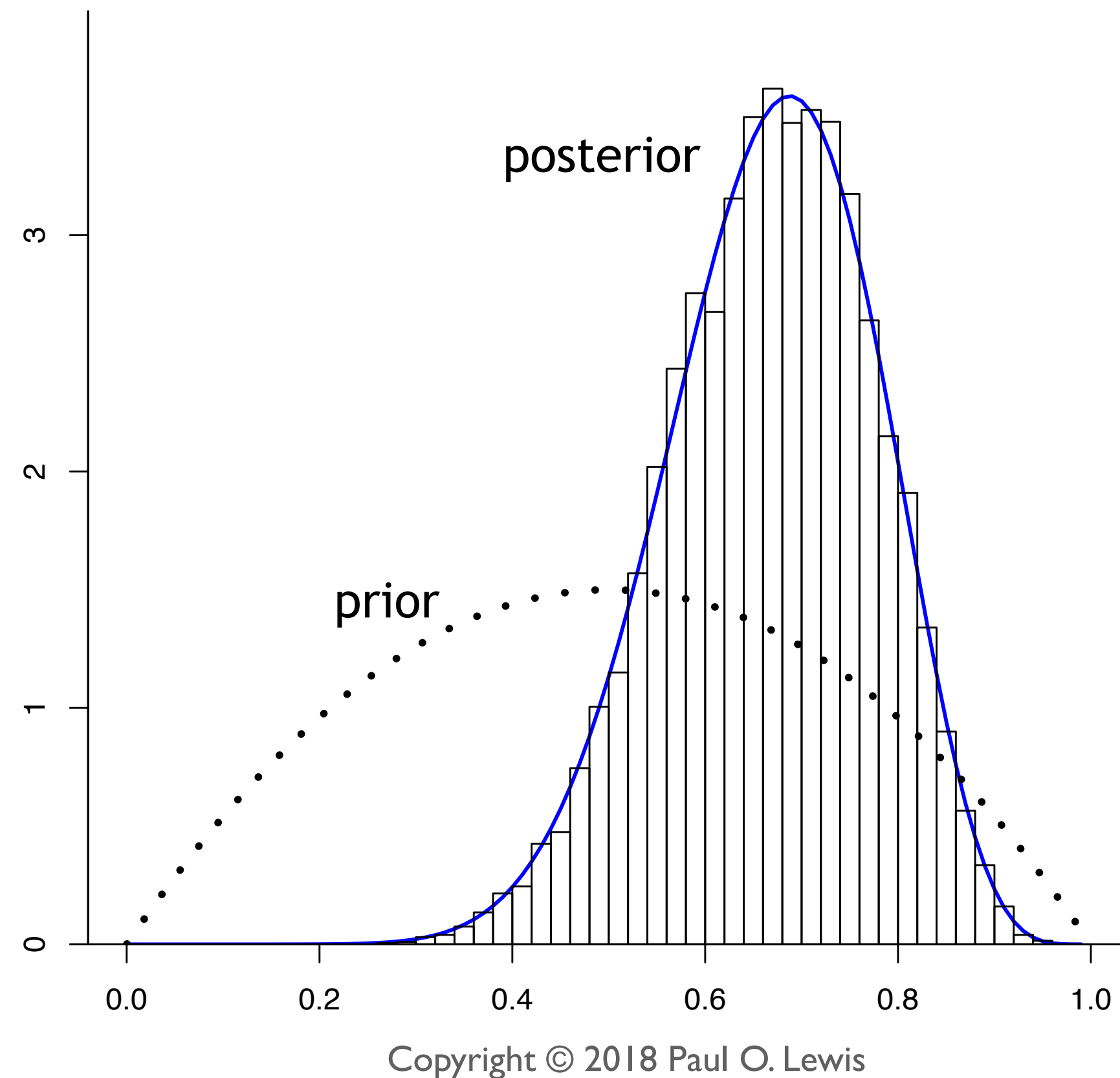
A group of algorithms for approximating the posterior distribution (also known as samplers)

Markov chain means the progress of the algorithm doesn't depend on its past

Monte Carlo (named for the casino in Monaco) methods estimate a distribution via random sampling

We use this algorithm to visit different regions the parameter space. The number of times a given region is visited will be in proportion to its posterior probability

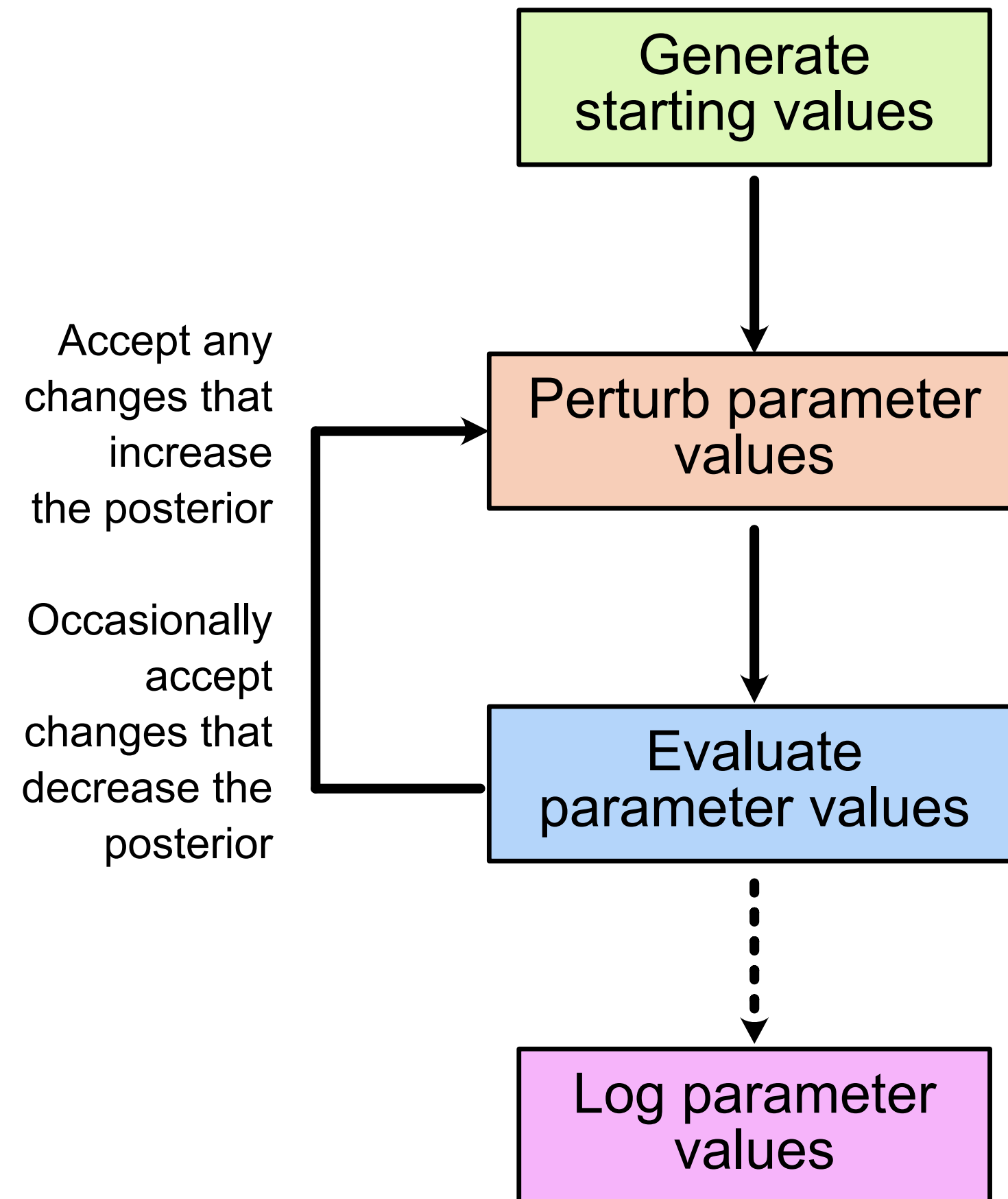
What is Markov chain Monte Carlo (MCMC)?



The aim is to produce a **histogram** that provides a good **approximation** of the posterior

The Metropolis-Hastings algorithm

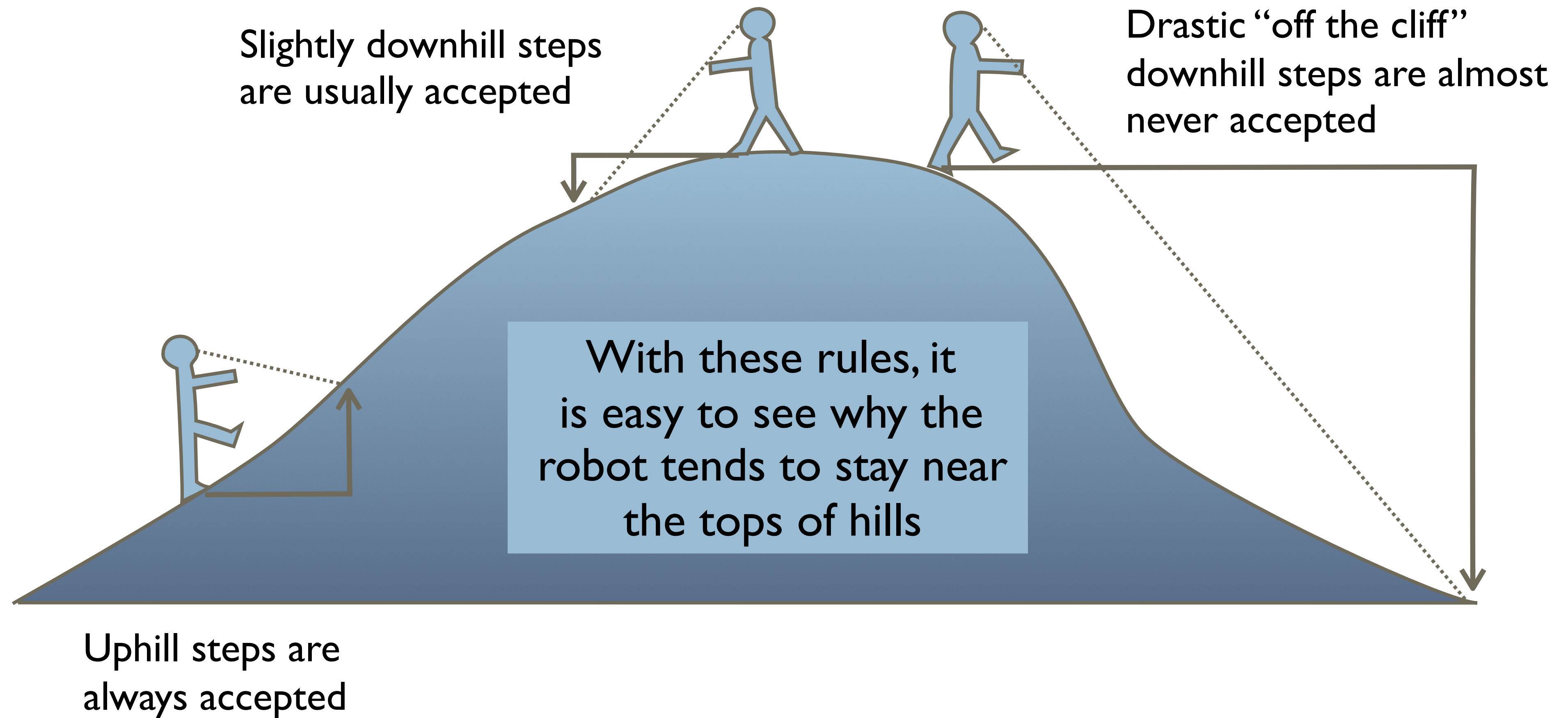
Flowchart

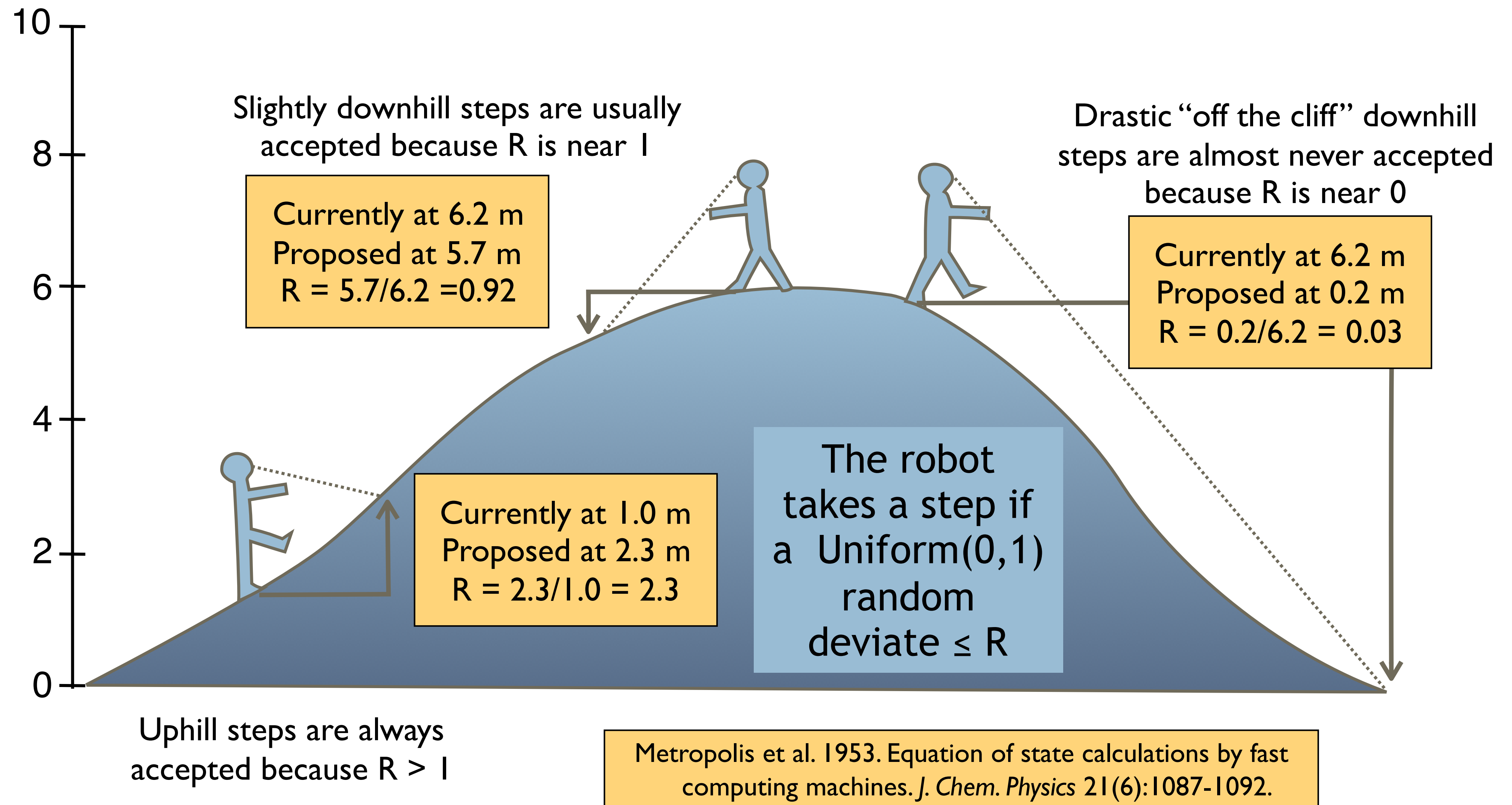


Pseudocode

```
initialize starting values;  
  
for i in mcmc steps  
do  
    propose new parameter values;  
    calculate the Hastings ratio R;  
  
    if( R > 1 )  
        accept the new values;  
    else  
        accept the new values with Pr = R;  
  
    store the values with frequency j;  
done
```

MCMC robot's rules





When calculating the ratio (R) of posterior densities, the marginal probability of the data cancels.

$$\frac{p(\theta^* | D)}{p(\theta | D)} = \frac{\frac{p(D | \theta^*) p(\theta^*)}{\cancel{p(D)}}}{\frac{p(D | \theta) p(\theta)}{\cancel{p(D)}}} = \frac{p(D | \theta^*) p(\theta^*)}{p(D | \theta) p(\theta)}$$

Posterior
odds

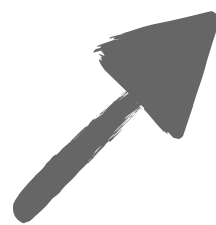
Apply Bayes' rule to
both top and bottom

Likelihood
ratio

Prior
odds

Hastings ratio

new parameter values



$$R = \frac{P(\text{HMM diagram with } * \mid \begin{smallmatrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{smallmatrix})}{P(\text{HMM diagram} \mid \begin{smallmatrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{smallmatrix})}$$

=

$$\frac{P(\begin{smallmatrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{smallmatrix} \mid \text{HMM diagram with } *) P(\text{HMM diagram with } *)}{P(\begin{smallmatrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{smallmatrix})}$$

The marginal probability of the data cancels out

$$\frac{P(\begin{smallmatrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{smallmatrix} \mid \text{HMM diagram}) P(\text{HMM diagram})}{P(\begin{smallmatrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{smallmatrix})}$$

=

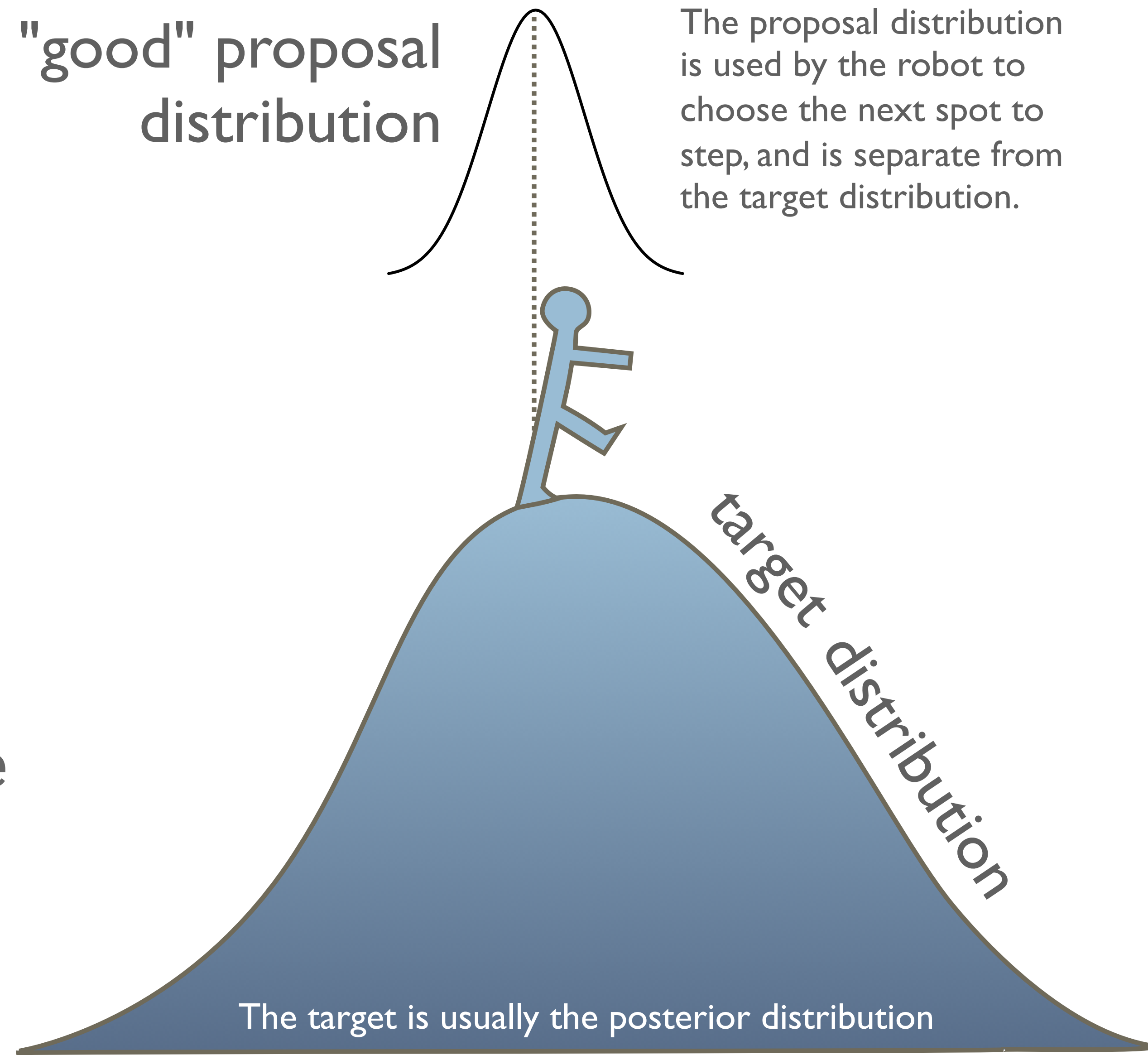
$$\frac{P(\begin{smallmatrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{smallmatrix} \mid \text{HMM diagram with } *) P(\text{HMM diagram with } *)}{P(\begin{smallmatrix} 0101\dots \\ 1101\dots \\ 0100\dots \end{smallmatrix} \mid \text{HMM diagram}) P(\text{HMM diagram})}$$

All we're left to calculate is the likelihood ratio and the prior odds ratio

Moves

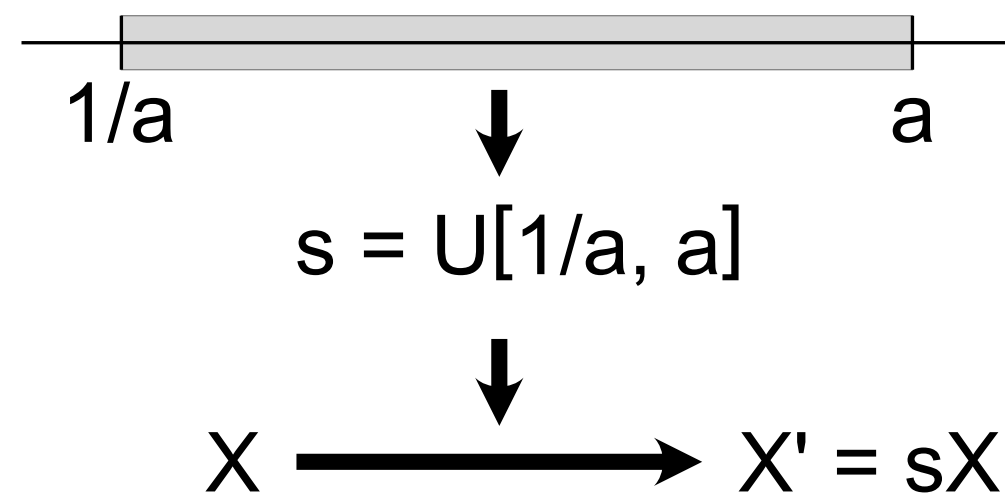
Known as **moves**,
proposals or **operators** —
how we move across the
parameter space

i.e., how do we select the
next set of parameters to
evaluate

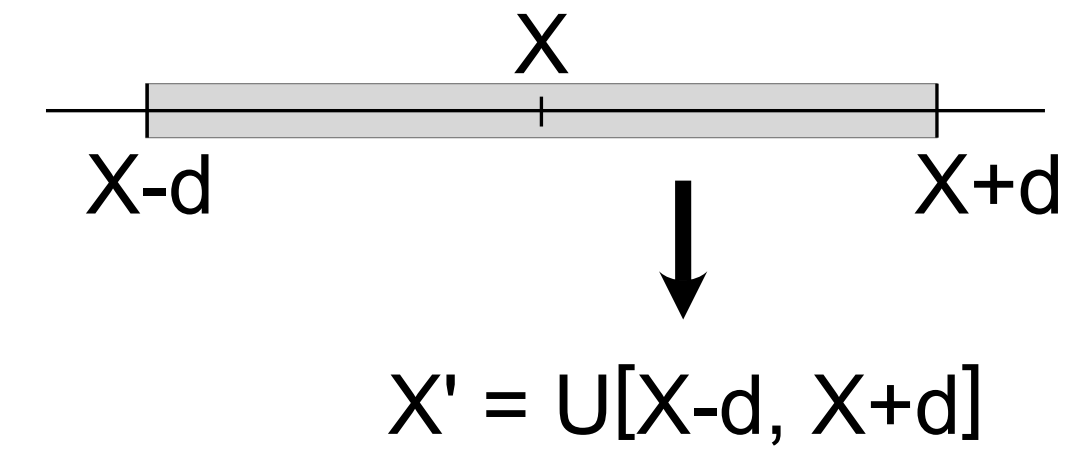


Moves

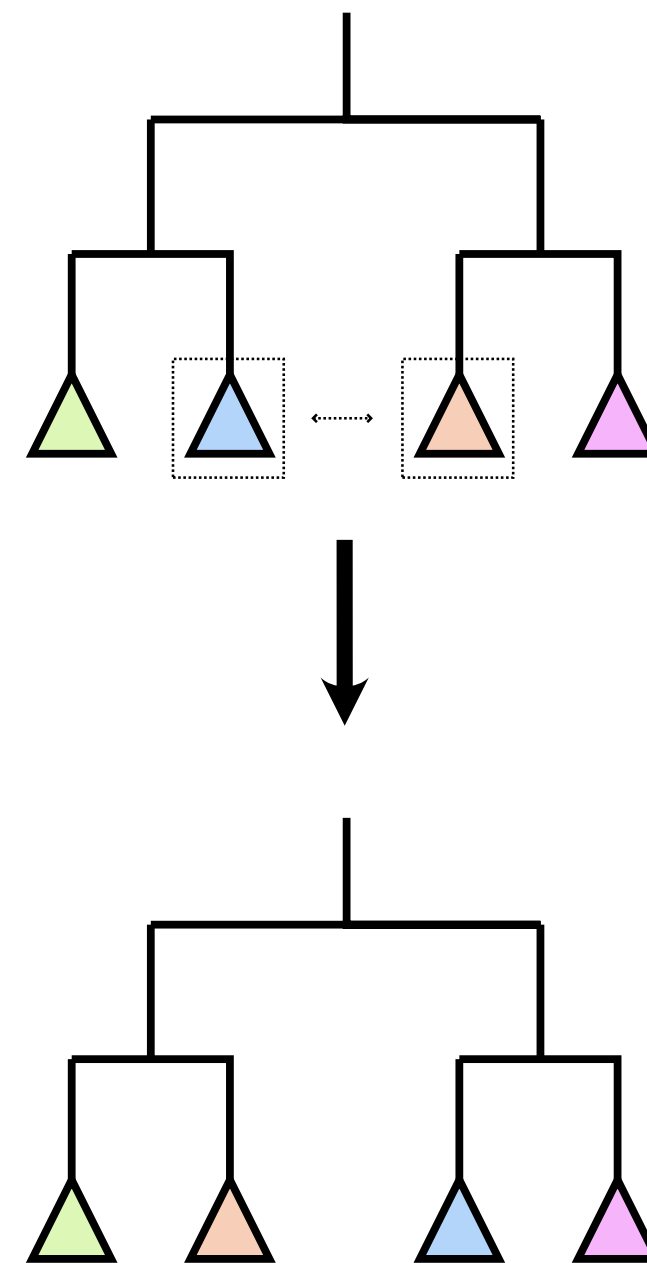
Scaling move



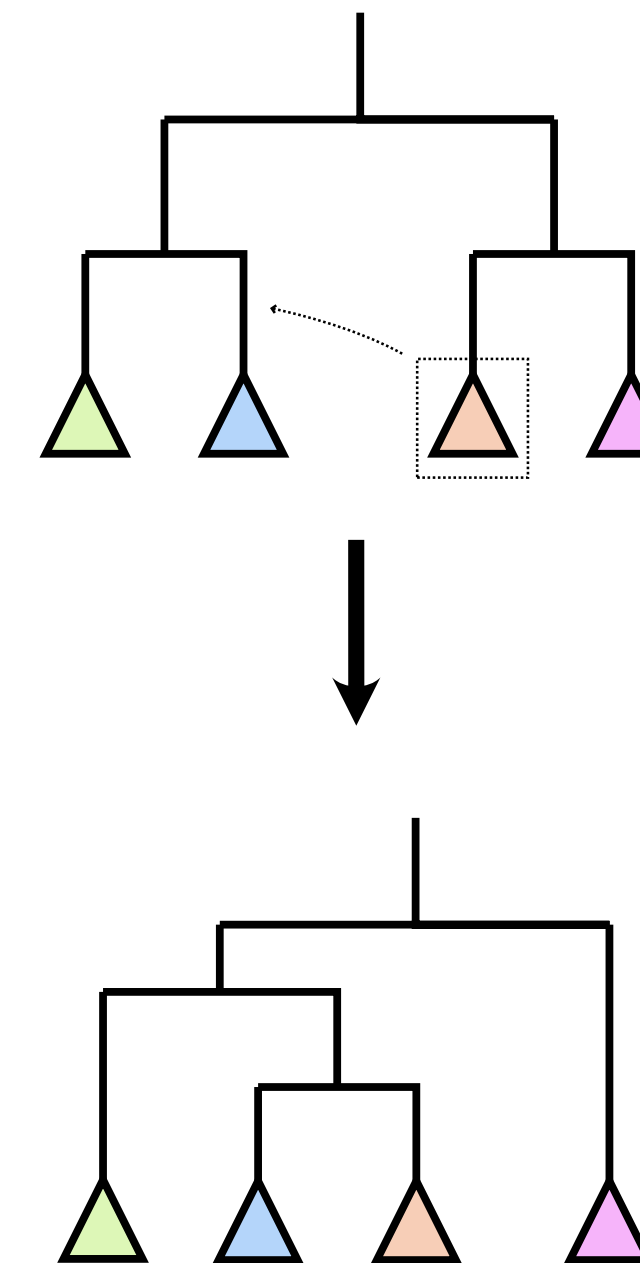
Sliding move



SubTree eXchange (STX) move



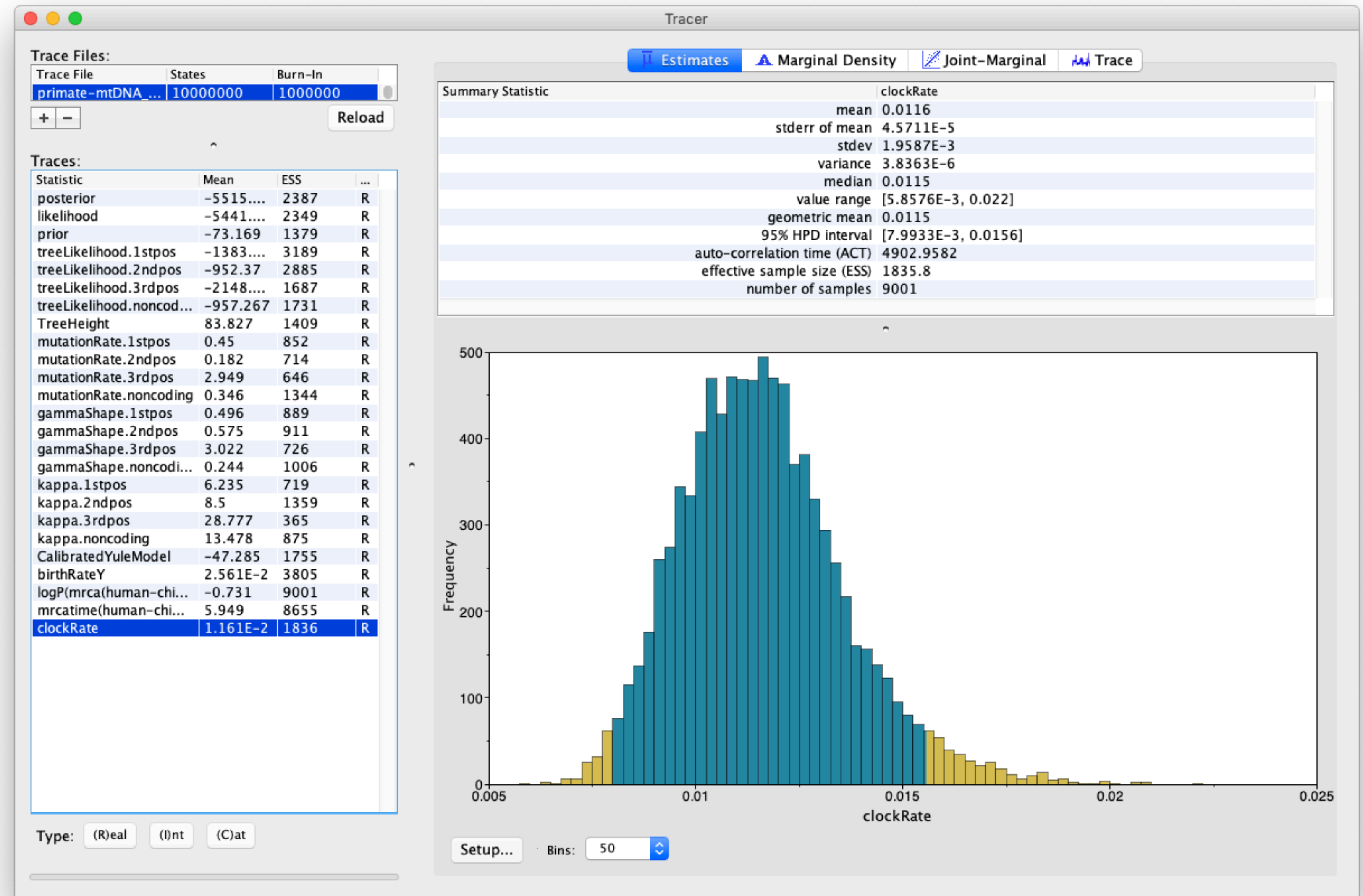
Subtree Pruning and Regrafting (SPR) move



MCMC Robot demo

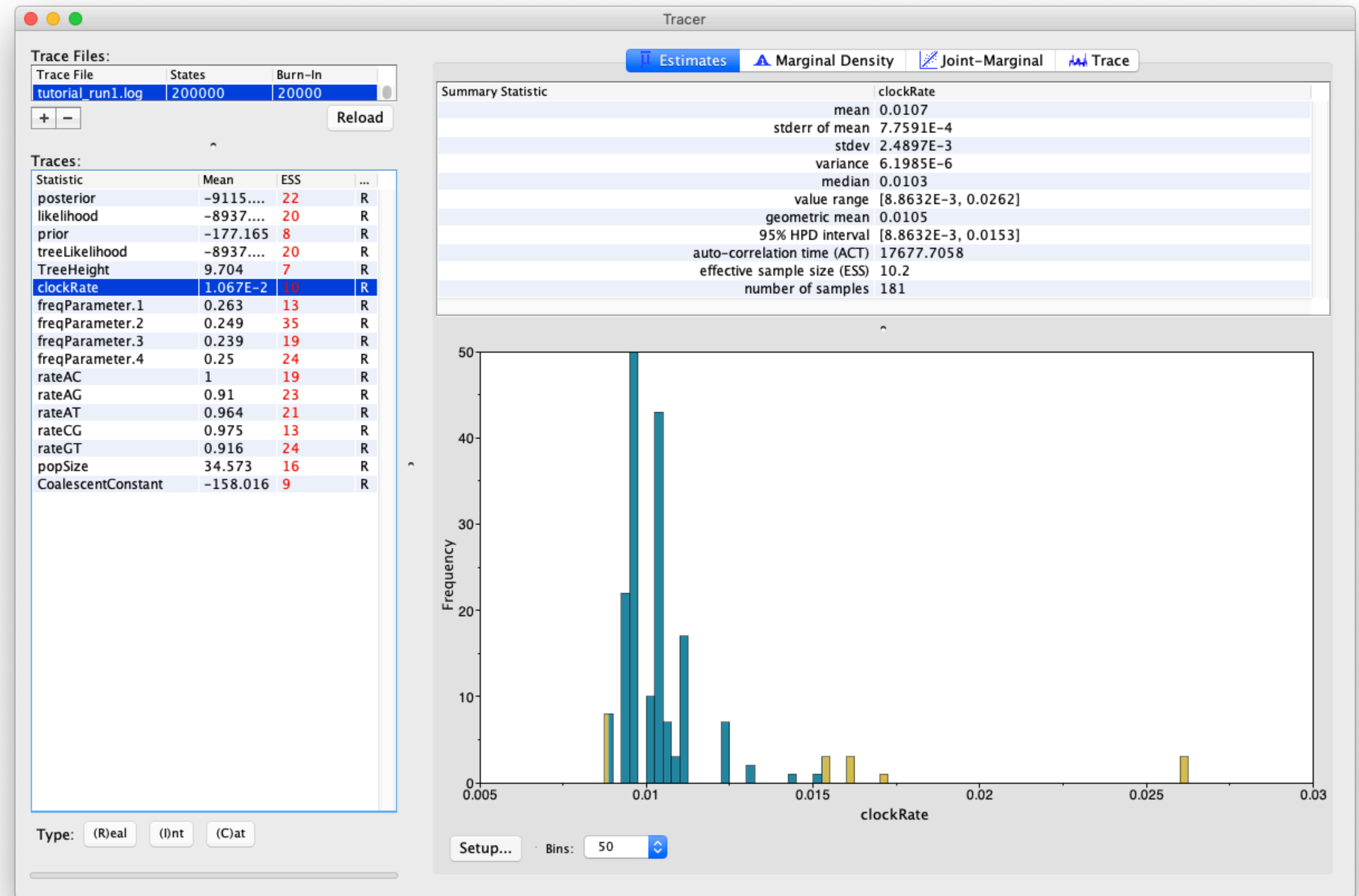
Summarising the posterior

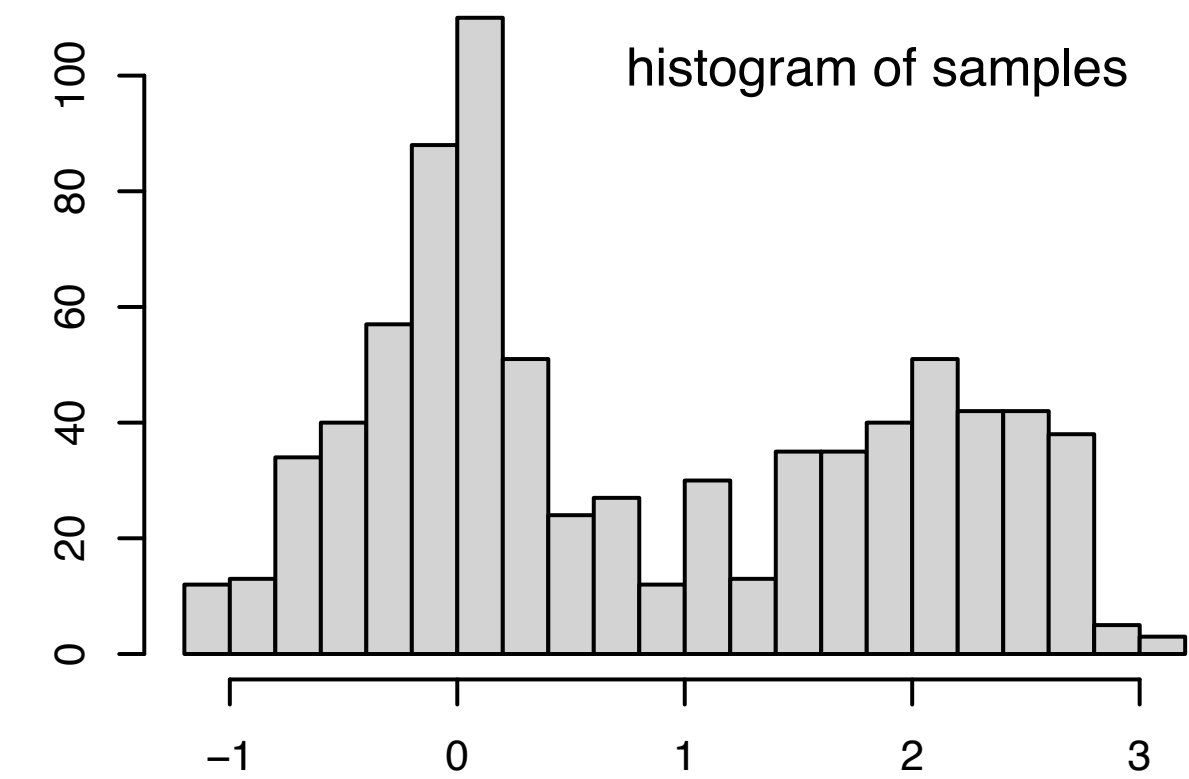
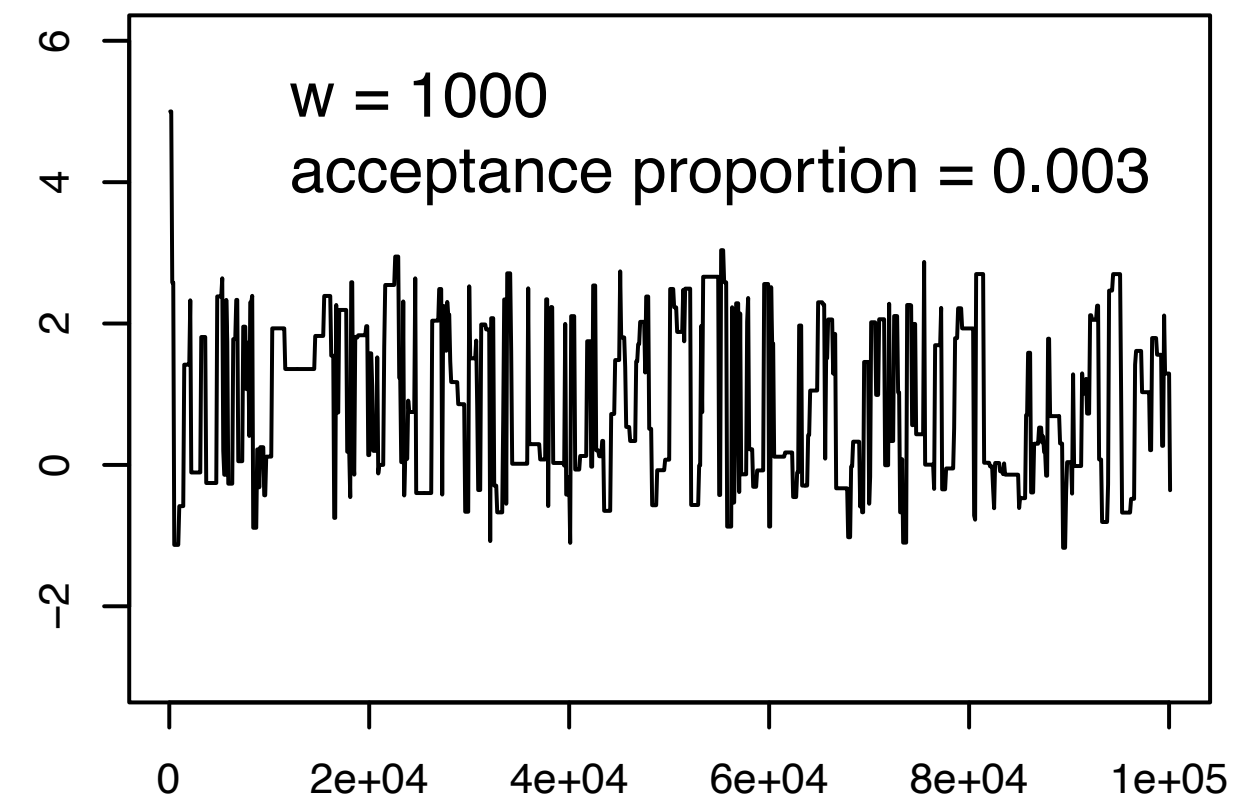
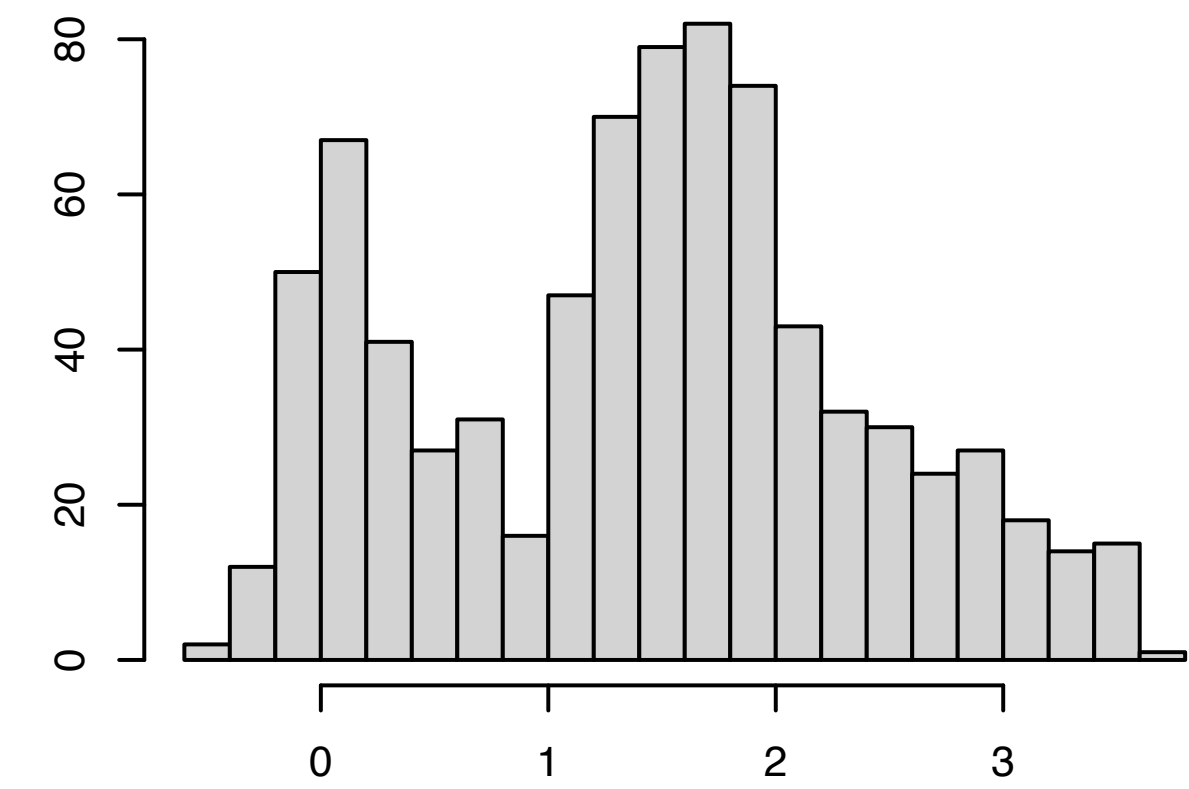
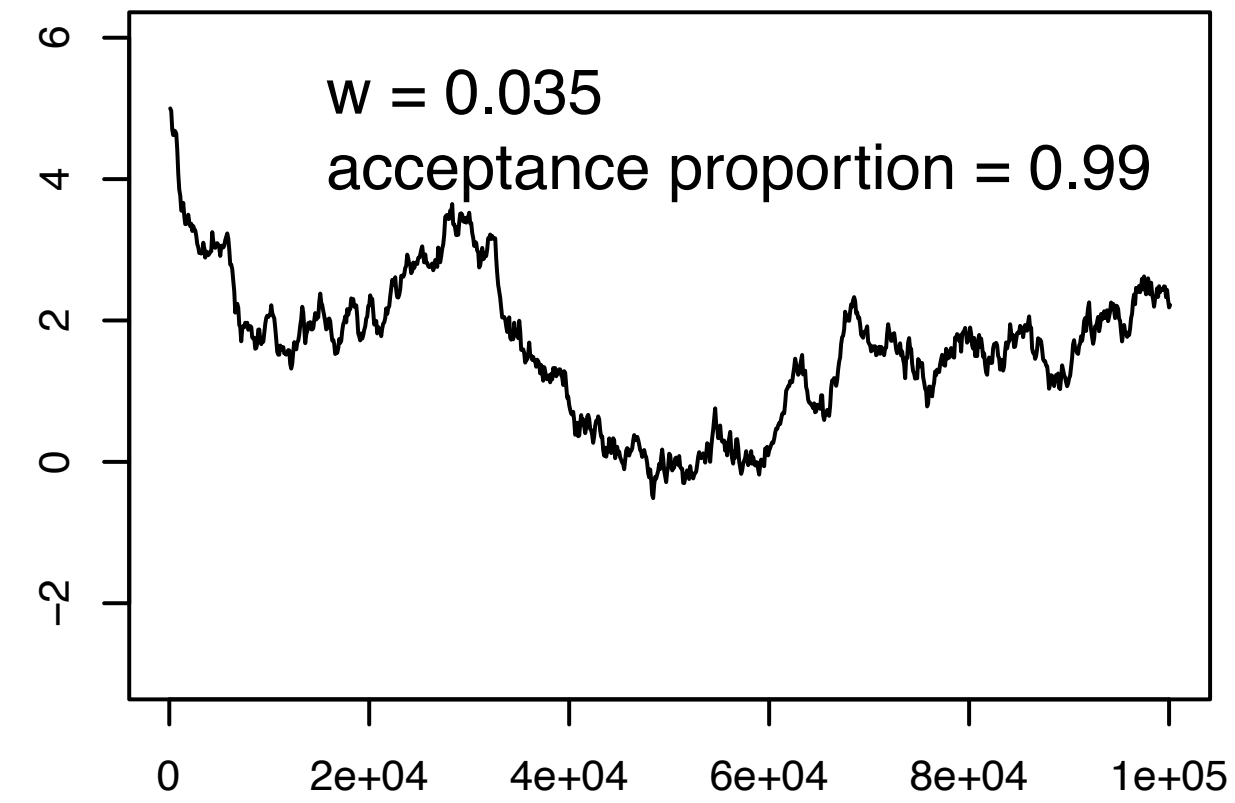
Tracer is an amazing program for exploring MCMC output



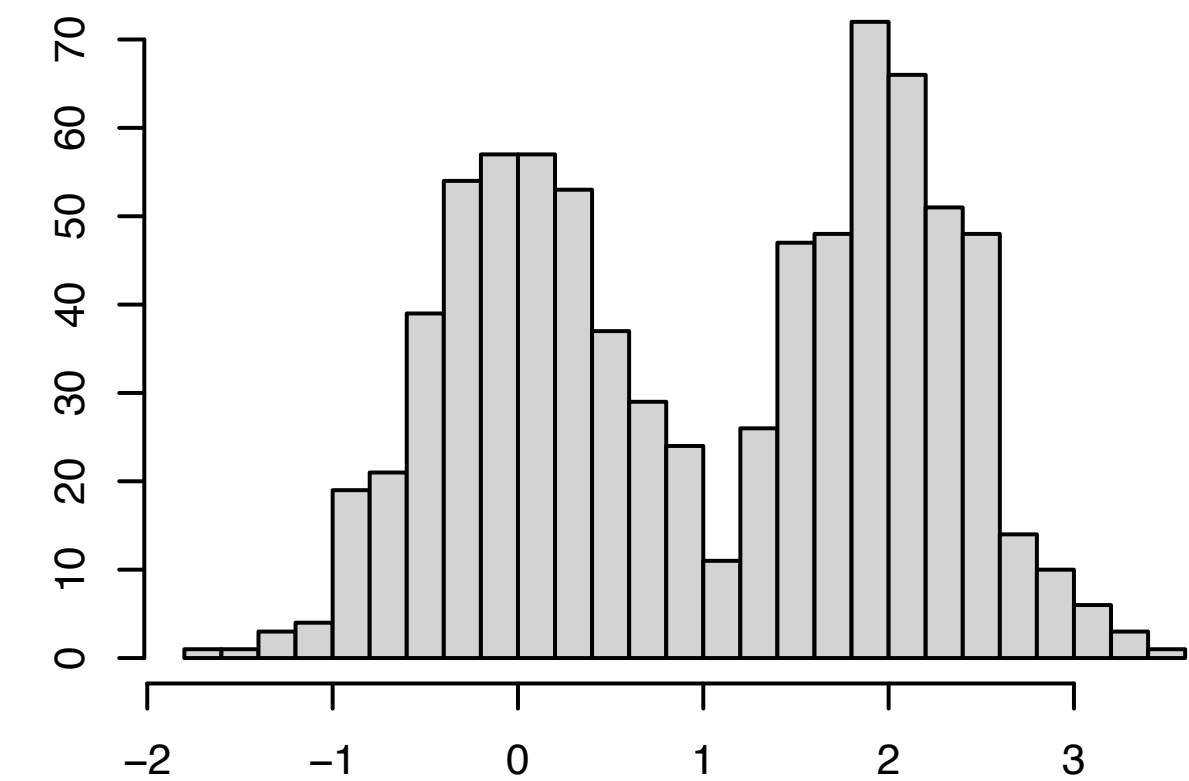
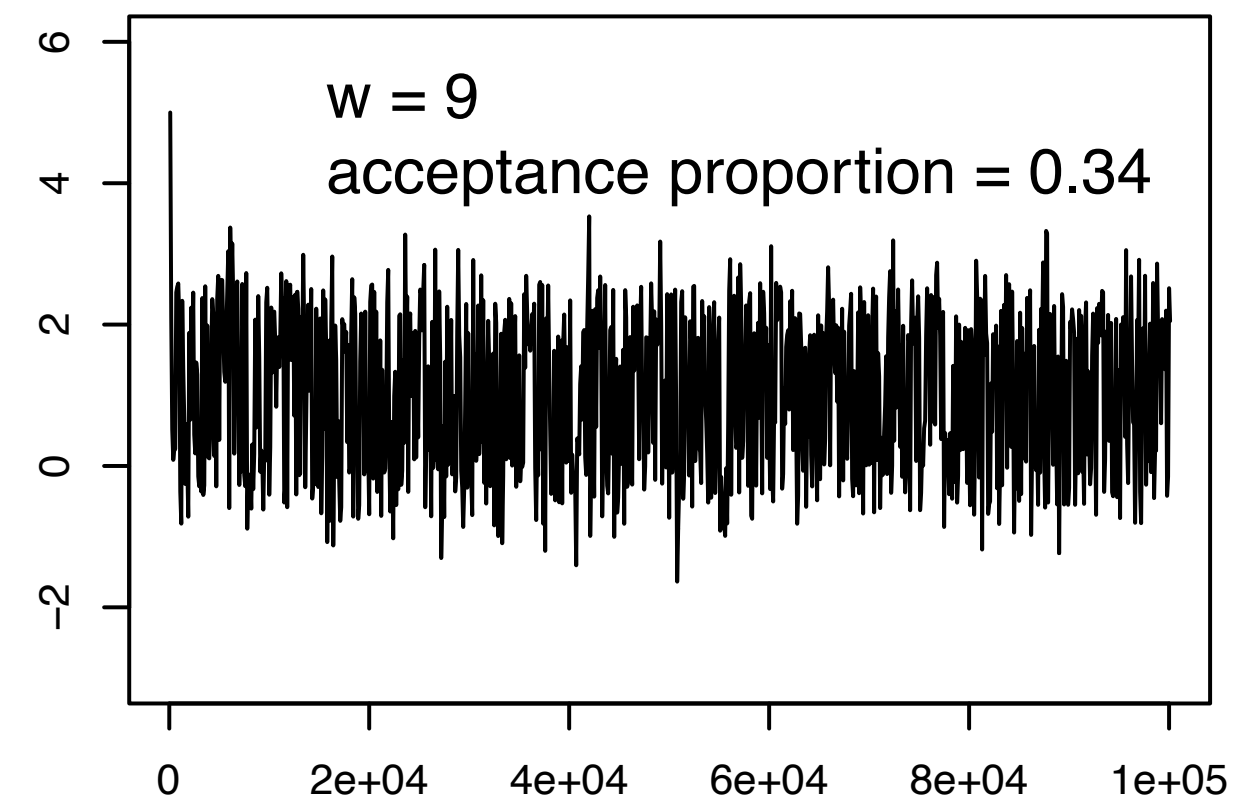
Summarising the posterior

Tracer is an amazing program for exploring MCMC output





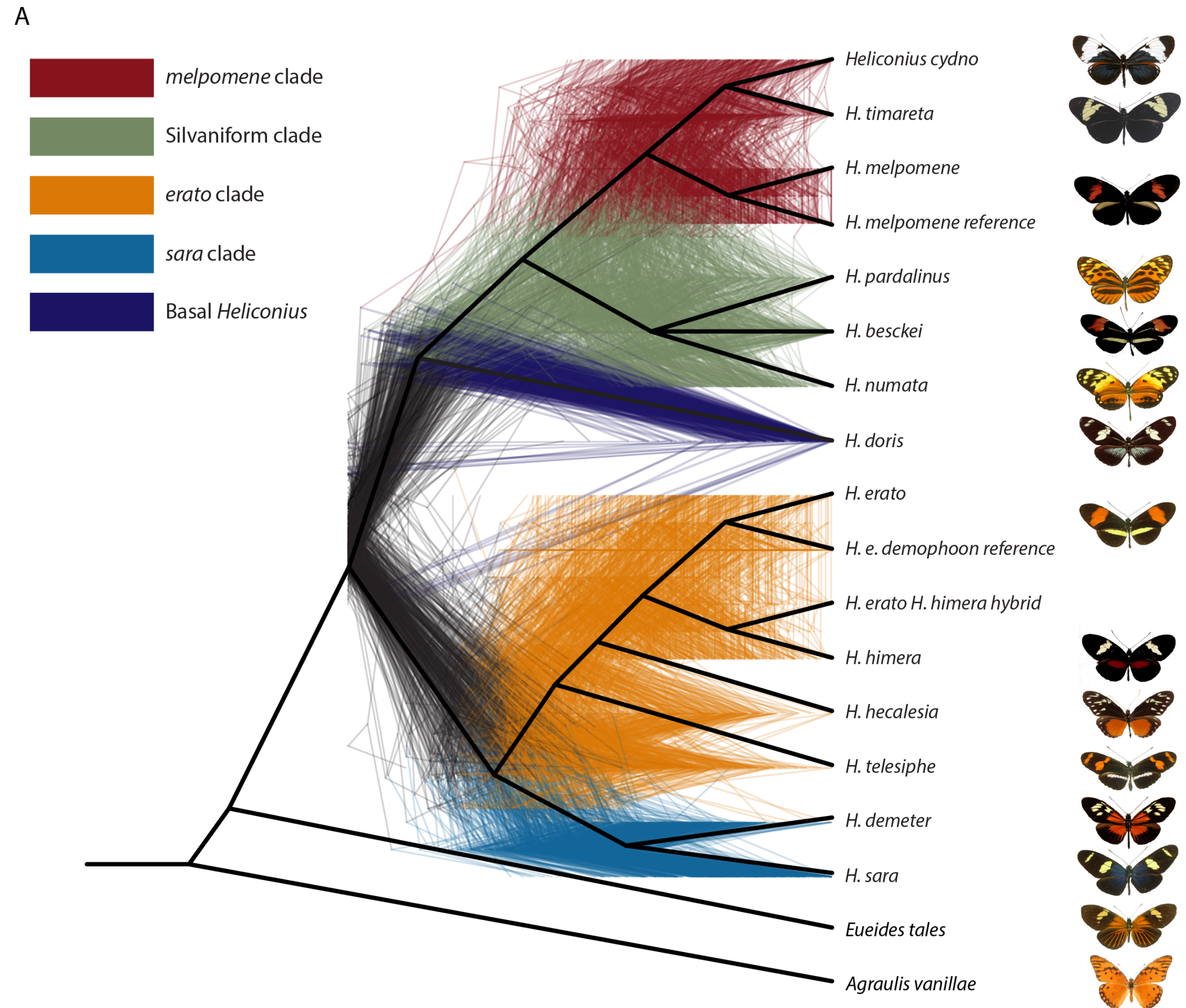
Better!



Summarising the posterior

Summarising trees is
much more challenging

Presenting a single
summary tree can be
misleading



Summarising the posterior

Maximum clade credibility (MCC) tree — the tree in the posterior sample that has the highest posterior probability (i.e., clade support) across all nodes

The **95% highest posterior density (HPD)** — the shortest interval that contains 95% of the posterior probability. The Bayesian equivalent of the 95% confidence interval

Marginal posterior density — the probability of a parameter regardless of the value of the others, represented by the histogram

MrBayes



The first widely used software for Bayesian phylogenetics

Well designed, incorporates a wide range of models

Not under (much) active development these days but remains very popular

Caveat: its not as flexible as some alternative options

Predecessor of RevBayes

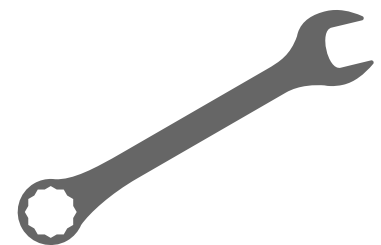
[MrBayes website](http://mrbayes.cit.uci.edu/)

Exercise

Reading



Bayesian Inference of Phylogeny and Its Impact on Evolutionary Biology *Huelsenbeck et al.* ([2001](#))



[MCMC Robot App](#) by Paul Lewis



For later: Practical guidelines for Bayesian phylogenetic inference using Markov chain Monte Carlo (MCMC) *Barido-Sottani et al.* ([2024](#))